KINGDOM OF SAUDI ARABIA
MINISTRY OF EDUCATION – JAZAN UNIVERSITY
COLLEGE OF ENGINEERING & COMPUTER SCIENCE

■ JAZAN UNIVERSITY ■
College of Engineering
& Computer Science

المملكة العربية السعودية
وزارة التعليم - جامعة جازان
كلية الهندسة وعلوم الحاسب

FINAL LAB EXAM

BACHELOR IN INFORMATION TECHNOLOGY

Academic Year: 2025 – 2026 Term: ● First /■ (Second)

Student Name: ________________________________	**Student ID:** _______________
Section Number: 14611	**Exam Date:** 20/05/2026
Course Name: Database Concepts and Design	**Exam Time:**
Course Code: INFS-211 **Course Level:** 3/4 **Total Marks:**	**10**

Marks Summary (10)

Question #	Performance Indicator	Total Marks	Obtained Marks
Q1	PI – (2.3)	10	

Teacher's Signature **Hussein Rajab** **10**

CLO-PI-SO Mapping

CLO IDs	SO-1	SO-2	SO-3	SO-4	SO-5	SO-6
CLO#01	PI 1.1	-	-	-	-	-
CLO#02	PI 1.3	-	-	-	-	-
CLO#03	-	PI 2.1	-	-	-	-
CLO#04	-	PI 2.3	-	-	-	-
CLO#05	PI 3.1					
CLO#06	PI 3.2					

Instructions for Students:

1. **Write your name and student ID in the giving Form**
2. **The questions from Ch5**

Submit the answer as a word document

Q1. Answer **the questions** of the following questions. **Question 1.1(a) and 1.1(b) is compulsory.**

Figure 6.1
Results of SELECT and PROJECT operations. (a) $\sigma_{\text{(Dno=4 AND Salary>25000) OR (Dno=5 AND Salary>30000)}}$(EMPLOYEE). (b) $\pi_{\text{Lname, Fname, Salary}}$(EMPLOYEE). (c) $\pi_{\text{Sex, Salary}}$(EMPLOYEE).

EMPLOYEE Relation (Reference Table):

Fname	Minit	Lname	Ssn	Bdate	Address	Sex	Salary	Super_ssn	Dno
John	B	Smith	123456789	09-Jan-65	731 Fondren, Houston TX	M	30,000	333445555	5
Franklin	T	Wong	333445555	08-Dec-55	638 Voss, Houston TX	M	40,000	888665555	5
Alicia	J	Zelaya	999887777	19-Jan-68	3321 Castle, Spring TX	F	25,000	987654321	4
Jennifer	S	Wallace	987654321	20-Jun-41	291 Berry, Bellaire TX	F	43,000	888665555	4
Ramesh	K	Narayan	666884444	15-Sep-62	975 Fire Oak, Humble TX	M	38,000	333445555	5
Joyce	A	English	453453453	31-Jul-72	5631 Rice, Houston TX	F	25,000	333445555	5
Ahmad	V	Jabbar	987987987	29-Mar-69	980 Dallas, Houston TX	M	25,000	987654321	4
James	E	Borg	888665555	10-Nov-37	450 Stone, Houston TX	M	55,000	NULL	1

1.1 Use above relation EMPLOYEE and write the relational algebra query.

a. Retrieve the records from EMPLOYEE whose department is 4.

Answer (a):

We apply the **SELECT operation (σ)** to the EMPLOYEE relation with the condition **Dno = 4**. This returns all tuples where the department number equals 4, preserving all 10 columns.

$$\sigma_{\text{Dno} = 4}\ (\text{EMPLOYEE})$$

Result — 3 tuples returned (employees in Department 4):

Fname	Minit	Lname	Ssn	Bdate	Address	Sex	Salary	Super_ssn	Dno
Alicia	J	Zelaya	999887777	19-Jan-68	3321 Castle, Spring TX	F	25,000	987654321	4
Jennifer	S	Wallace	987654321	20-Jun-41	291 Berry, Bellaire TX	F	43,000	888665555	4
Ahmad	V	Jabbar	987987987	29-Mar-69	980 Dallas, Houston TX	M	25,000	987654321	4

b. Select the tuples where salary > 40000.

Answer (b):

We apply the **SELECT operation (σ)** to the EMPLOYEE relation with the condition **Salary > 40000**. This returns every tuple whose Salary is strictly greater than 40,000, preserving all 10 columns.

$$\sigma_{\text{Salary} > 40000}\ (\text{EMPLOYEE})$$

Result — 2 tuples returned (employees with Salary > 40,000):

Fname	Minit	Lname	Ssn	Bdate	Address	Sex	Salary	Super_ssn	Dno
Jennifer	S	Wallace	987654321	20-Jun-41	291 Berry, Bellaire TX	F	43,000	888665555	4
James	E	Borg	888665555	10-Nov-37	450 Stone, Houston TX	M	55,000	NULL	1